

Electrochemical CO₂ Reduction over Metal-/Nitrogen-Doped Graphene Single-Atom Catalysts Modeled Using the Grand-Canonical Density Functional Theory

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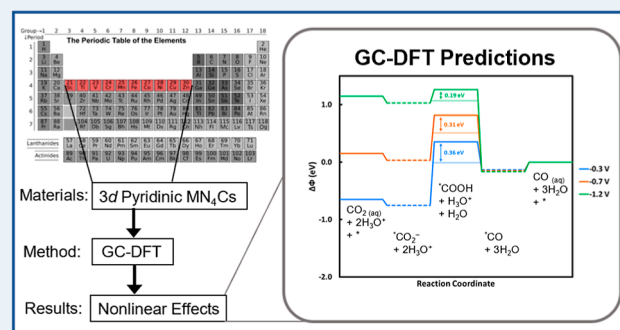
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ABSTRACT: Renewably driven, electrochemical conversion of carbon dioxide into value-added products is expected to be a critical tool in global decarbonization. However, theoretical studies based on the computational hydrogen electrode largely ignore the nonlinear effects of the applied potential on the calculated results, leading to inaccurate predictions of catalytic behavior or mechanistic pathways. Here, we use grand canonical density functional theory (GC-DFT) to model electrochemical CO₂ reduction (CO₂R) over metal- and nitrogen-doped graphene catalysts (MNCs) and explicitly include the effects of the applied potential. We used GC-DFT to compute the CO₂ to CO reaction intermediate energies at −0.3, −0.7, and −1.2 V_{SHE} catalyzed by MNCs each doped with 1 of the 10 3d block metals coordinated by four pyridinic nitrogen atoms. Our results predict that Sc-, Ti-, Co-, Cu-, and Zn-N₄Cs effectively catalyze CO₂R at moderate to large reducing potentials (−0.7 to −1.2 V_{SHE}). ZnN₄C is a particularly promising electrocatalyst for CO₂R to CO both at low and moderate applied potentials based on our thermodynamic analysis. Our findings also explain the observed pH independence of CO production over FeN₄C and predict that the rate-determining step of CO₂R over FeN₄C is not *CO₂[−] formation but rather *CO desorption. Additionally, the GC-DFT-computed density of states analysis illustrates how the electronic states of MNCs and adsorbates change non-uniformly with applied potential, resulting in a significantly increased *CO₂[−] stability relative to other intermediates and demonstrating that the formation of the adsorbed *CO₂[−] anion is critical to CO₂R activation. This work demonstrates how GC-DFT paves the way for physically realistic and accurate theoretical simulations of reacting electrochemical systems.

KEYWORDS: single-atom catalysts, electrochemistry, grand canonical density functional theory, DFT calculations, surface chemistry, CO₂ reduction



INTRODUCTION

Excess atmospheric carbon dioxide (CO₂) is a major contributor to global warming, yet CO₂ is emitted from nearly every modern industrial process. As an alternative to atmospheric release or sequestration, electrochemical CO₂ reduction (CO₂R) can convert captured CO₂ into value-added products.¹ CO₂R on pure metal electrodes, such as copper, requires large overpotentials to drive CO₂ conversion and competes with the hydrogen evolution reaction (HER) at all potentials.^{2–4} Gold and silver, which are arguably the best electrocatalysts for CO₂R to CO, are prohibitively expensive for large-scale applications.^{5,6}

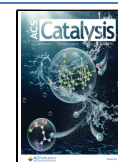
An emergent class of electrocatalysts for CO₂R is two-dimensional metal-nitrogen-doped carbon (MNC) materials where a metal center, usually a first-row transition metal, is bound to the nitrogen atoms of N-doped graphene.⁷ The metal atoms of MNCs are assumed to be the catalytically active sites

and are analogous to those of naturally occurring porphyrin complexes, such as chlorophyll and heme.⁸ MNCs are an attractive alternative to bulk metal catalysts as they have near maximal metal atom utilization, are highly selective for reducing CO₂ to CO at moderate potentials, and are composed of relatively earth abundant and inexpensive elements.^{2,9} Experimental studies have demonstrated the high selectivity of CO₂R catalyzed by several MNCs toward CO with MnN_xC (where typically $x \in [1, 2, 3, 4]$) achieving a Faradaic efficiency (FE) of up to 80%, FeN_xC and NiN_xC

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reaching FEs over 85%, and ZnN_xC attaining a FE of up to 95%.^{10–17}

While MNCs have exhibited significant promise for CO_2R , many questions persist about the nature of their active sites and activation mechanisms. MNC synthesis generates a range of dispersed atomic structures that likely contribute unequally to the catalytic activity and promote different catalytic mechanisms. The MN_4C sites are observed to have a square-planar geometry, where the pyridinic coordination of the metal by four nitrogen atoms (Figure 1) is reported to be the most

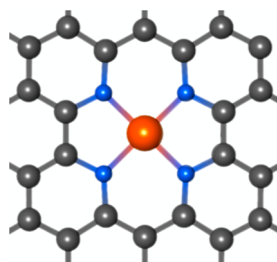


Figure 1. Structure of an MN_4C catalyst where a 3d-transition metal atom (orange) is coordinated to four pyridinic nitrogen atoms (blue) doped into a graphene lattice (gray).

stable, although recent work has strongly suggested that a lower nitrogen coordination might increase the catalytic activity of CO_2R over some MNCs.^{17–20} Ideally, future materials for CO_2R catalysis would synthetically select for the most active site; however, experimental determination of the CO_2R active site remains a daunting challenge even with state-of-the-art electrochemical characterization techniques.²¹ In contrast, computational models can be used to directly investigate the reactivity of hypothetical reactive sites and mechanisms as they explicitly model reactions in atomic detail. For this work, we developed atomistic models that describe the CO_2R activity of the pyridinic MN_4C site because it is the most stable MNC configuration, the most widely reported active site for MNC-catalyzed CO_2R to CO, and highly selective toward CO production through the suppression of the HER.^{22–24}

The number of electrons at a reactive site of an electrochemical interface depends on the applied potential, catalyst, solvent, electrolyte, and state along the reaction potential energy surface. However, modeling the electrified catalyst–solvent interface with traditional density functional theory (DFT) is not physically realistic because it is a canonical, fixed-electron number approach (which we refer to as unbiased calculations).²⁵ For over a decade, the simple and elegant computational hydrogen electrode (CHE) model has served as the workhorse for estimating the effects of applied potential on electrocatalytic processes.²⁶ However, the tradeoff for its ease of application is that it does not explicitly include the electrode potential in the quantum chemical model and instead adjusts for the potential through post-minimization algebraic corrections. DFT has recently been extended to grand-canonical DFT (GC-DFT), which naturally accounts for the applied potential by self-consistently varying the number of electrons in the system such that its Fermi level is in equilibrium with the applied external potential.^{27,28}

Additionally, the constraint of charge neutrality requires the use of a solvent model in GC-DFT calculations to account for the variable number of electrons.²⁹ DFT has traditionally been

used with a catalyst–vacuum interface model to describe catalyst–electrolyte interfaces;^{12,30–32} however, several researchers have approximated the solvation energies of intermediates and added them as empirical correction values,^{31,33–35} which provided more accurate results than model vacuum calculations. More recently, solvation effects have been modeled with a polarizable continuum and/or explicit solvent molecules,³⁶ both producing moderately more accurate energies than previous approaches. Our group has previously shown, and further demonstrates in this work, that surface and adsorbate geometries can vary significantly with applied potential and solvent model, resulting in Helmholtz free energies and a projected density of states (PDOS) that change non-linearly with bias, which indicates that orbital mixing is significantly affected by the local electric field.²⁵ The results of these works have shown that GC-DFT is able to capture CO_2R behavior that the CHE + an implicit solvent model cannot, both for well-studied systems (such as Cu and Ag) and for novel systems. Thus, the major advantage of GC-DFT is its ability to self-consistently account for both the interdependent effects of the solvent and the applied bias.

In this contribution, we report the results of applying GC-DFT to the electrocatalytic reduction of CO_2 to CO over pyridinic MN_4Cs formed by the 10 3d-block metals. We also provide a fundamental description of the effects of the applied potential on the electronic structure of CO_2 adsorbed on pyridinic MN_4C sites through PDOS analysis. Although different nitrogen functionalities (pyridinic, pyrrolic, etc.), defect sites, electrolyte choice, and the sensitivity of activation barriers to the potential affect the reactivity of active sites toward CO_2R , they are beyond the scope of the present study. To the best of our knowledge, this is the first study to fully implement GC-DFT to comprehensively compare the thermodynamics of the 10 3d-block MN_4Cs in the pyridinic configuration for catalyzing the electrochemical CO_2R reaction. This work also illustrates how GC-DFT is a viable new approach for accurately and realistically modeling electrocatalytic systems.

METHODS

Method Justification. Most DFT studies of electrochemical processes calculate the reaction energetics using conventional DFT corresponding to an unbiased, neutral charge system. Such studies typically treat solvation using implicit or explicit models and account for applied bias effects using empirical corrections from the CHE.^{7,9,12,28,37–39} The CHE approach adjusts the DFT-computed energies of adsorbed species by a constant value of $ne\phi$, where ϕ is the potential and n is the formal number of electrons transferred by the redox process, and is restricted to only estimating energies of intermediates that are formed via a proton-coupled electron transfer (PCET) mechanism. Furthermore, the CHE model neglects the direct effects of the applied bias on the interface, including the change in the electronic structure caused by the addition or removal of electrons, the non-integer changes in transferred electrons, the change in the electronic states because of polarization by the applied electric field, and the change in the geometry of the reacting species at different applied biases.²⁵

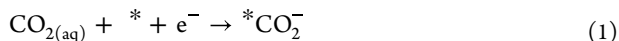
To describe the interactions between adsorbates and the electrified interface, the grand free energy can be computed within the grand canonical ensemble formalism in which the temperature, volume, and electron chemical potential (Fermi

energy) are fixed. A constant electron chemical potential dictates that the number of electrons is self-consistently varied and thus the GC ensemble is a more natural ensemble with which to calculate properties of electrochemical systems.^{25,28} GC-DFT is an ab initio approach that computes the grand free energy by self-consistently solving the Kohn–Sham equations while varying the number of electrons at the electrode to maintain a constant Fermi level that corresponds to the applied potential. Although GC-DFT calculations are more computationally demanding, when coupled with proper solvation models, they more accurately describe states along the reaction coordinate as a function of the potential which, in many cases, differ considerably from the results of conventional DFT calculations.^{25,28,29,40} For a more theoretical description of GC-DFT and its application to electrochemistry, the reader is referred to Sundararaman et al.'s work.²⁸

Computational Details. The intermediate geometries along the CO₂R reaction pathway on MN₄Cs were optimized using GC-DFT as implemented within Sundararaman et al.'s open-source JDFTx code.²⁷ MN₄C reactive sites were modeled with a 12 Å × 12 Å × 22 Å 2-dimensional supercell comprising 44 carbon atoms, 4 pyridinic nitrogen atoms, and 1 3d metal center as well as implicit solvent that fills the ~20 Å separation between MN₄C sheets. A planewave expansion with a 544 eV (20 Hartree) energy cutoff and a 3 × 3 × 1 gamma-centered *k*-point grid were confirmed to provide good approximations to the asymptotic energy and *k*-point limits, respectively. Geometry optimizations were performed for all adsorbates until the total energies and forces were converged to within 10^{−6} Ht/cell and 10^{−4} Ht/Bohr, respectively. All GC-DFT calculations were performed using the generalized gradient approximation Perdew–Burke–Ernzerhof (GGA-PBE) exchange correlation functional combined with Grimme's DFT-D3 van der Waals correction and the GBRV pseudopotentials.^{41,42} GGA-PBE was selected because of its high computational speed and relatively accurate prediction of free energies of heterogeneous catalysts. The DFT-D3 correction was chosen due to its structure dependence for calculating van der Waals interactions and its high accuracy in correcting for them. To describe the solvent, we chose the charge-asymmetric nonlocally determined local-electric (CANDLE) continuum solvation model due to its accurate description of polar solvents and their interactions with charged surfaces.⁴⁰ GC-DFT maintains charge neutrality of the periodic unit cell as the electron number changes with φ by introducing a compensating number of counterions in the electrolyte that distribute themselves to minimize the grand free energy.²⁷ One limitation of the CANDLE model is that it neglects ionic size effects on the reaction energetics, which have been shown to significantly alter catalytic activity for CO₂R.^{43–45} However, the effects of cation size on the CO₂R reaction are beyond the scope of this study. A more detailed description of GC-DFT, energy calculations, and the solvation settings used in this study are provided in the [Supporting Information](#).

RESULTS AND DISCUSSION

Several studies have suggested that a PCET is not involved in the rate-determining step (RDS) of CO₂R over MNCs or their molecular analogues.^{13,37,46,47} Therefore, we evaluated a mechanism that proceeds through a decoupled-proton-electron-transfer (DPET) pathway as described by eqs 1–4



The asterisk (*) indicates an open MN₄C adsorption site. Note that only the second and third steps involve a proton transfer. [Figure 2](#) shows representative intermediate structures

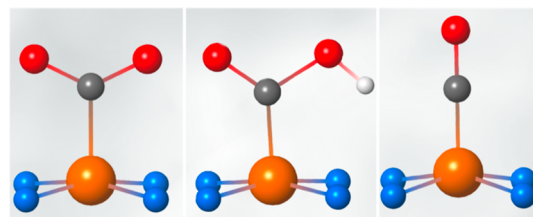


Figure 2. Geometries of CO₂R intermediates. The metal, carbon, nitrogen, oxygen, and hydrogen atoms are represented by orange, gray, blue, red, and white spheres, respectively.

along the CO₂R reaction coordinate. The grand free energies of these intermediates were computed at applied potentials of $\varphi = -0.3$, -0.7 , and -1.2 V versus the standard hydrogen electrode (herein referred to as V_{SHE}) and are plotted in [Figures 3–5](#). Transition state calculations are not performed in this study due to their significant computational cost within the GC-DFT approach. To connect the thermodynamics of our study with kinetics, we set a grand free energy threshold of 0.7 eV at which all catalysts with specific reaction steps more endergonic than this threshold were deemed too sluggish on lab time scales to be kinetically viable at 298 K. We note that this criterion is a necessary but insufficient condition for a mechanism to be kinetically active as the rate of a reaction path may be impractically slow because of transition states that lie too high in energy or less often because the activation entropy of the RDS is too low.

Our GC-DFT results show that CO₂ does not adsorb chemically to a few 3d MN₄C catalysts at the metal site at various applied potentials. In the cases of NiN₄C and CuN₄C at all potentials as well as ZnN₄C at $-0.3 V_{\text{SHE}}$ and $-0.7 V_{\text{SHE}}$, the first step of the mechanism involves a PCET, where aqueous CO₂ reacts with a proton-electron couple to form an adsorbed COOH. Note that this mechanism (i.e., PCET mechanism) is different from the one shown above (i.e., DPET mechanism). We discuss these cases below and provide more detail in the [Supporting Information](#).

The grand free energies of the CO₂R reaction on all 10 pyridinic MN₄C systems are shown in [Figures 3–5](#). A comparison of the GC-DFT calculated reaction coordinates and CHE calculated reaction coordinates are shown in [Supporting Information](#), [Figures S6–S8](#). All grand free energies were referenced to the product state, $\text{CO}_{(\text{aq})} + 3\text{H}_2\text{O}_{(\text{aq})} + *$. This choice makes the energies of the product states for each system 0.0 eV at all biases, and the energies of the preceding states, including the reactants ($\text{CO}_{2(\text{aq})} + 2\text{H}_3\text{O}^+_{(\text{aq})} + *$), all potential-dependent. However, the choice of the reference as either the reactant or product states is arbitrary as both reactants and products consist of solvated species and an open MN₄C site. Thus, the endergonicity or exergonicity of each reaction step and of the overall reaction is the same with either convention. We choose the products as

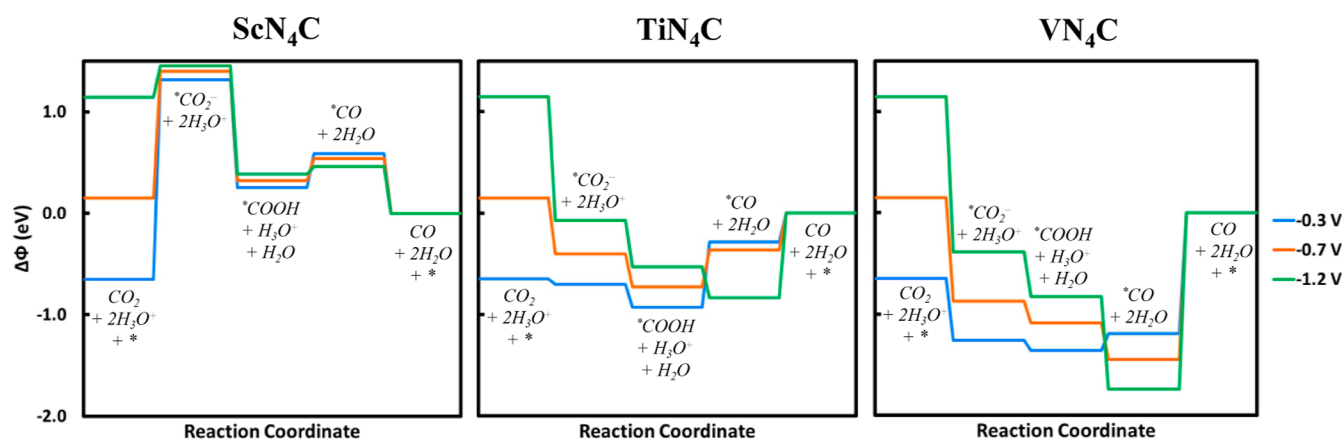


Figure 3. CO₂R reaction coordinate diagrams for Sc-, V-, and TiN₄C. Reaction grand free energies computed for applied reducing potentials of $\varphi = -0.3 V_{\text{SHE}}$ (blue), $-0.7 V_{\text{SHE}}$ (orange), and $-1.2 V_{\text{SHE}}$ (green). Note that the nonlinearity of energy changes with applied potential is common across all three catalysts.

the reference state to follow a conventional workflow as our products consist of neutral molecules and are hence unaffected by applied external bias, assuming the operating conditions are between the molecules' oxidation and reduction potentials.

The GC-DFT computed trends in the grand free energies with applied potential are unique to the adsorbed intermediate and transition metal of the MN₄C. GC-DFT predicts that the energies of the reactant state, $\text{CO}_{2(\text{aq})} + 2\text{H}_3\text{O}_{(\text{aq})}^+ + *$, change more rapidly with φ than those of adsorbed CO₂ on all 10 MN₄Cs. This indicates that the CO₂ adsorption step is critical in activating CO₂R, despite commonly being ignored in various conventional DFT studies.^{18,19,31,48,49} In this section, we divide our analysis into CO₂R over MN₄Cs composed of early, middle, and late 3d-block metals. We first discuss early 3d metal MN₄Cs.

CO₂R on Early 3d Metal MN₄Cs. There is a dearth of computational and experimental CO₂R studies of early 3d metals as dopants in single-atom catalysts because they are believed to be strong *OH and *CHO binders,^{7,50} which limit their selectivity toward CO production. Figure 3 shows the reaction coordinate diagrams for CO₂R on ScN₄C, TiN₄C, and VN₄C at three reducing potentials computed using GC-DFT. The computed reaction energetics reported in these diagrams show that the shifts in the intermediate energies are significant and non-linear, in contrast to the predictions of the CHE model. They also show that adsorbed CO₂R intermediates become more stable as the atomic number of the metal center increases, and that metal centers with higher atomic numbers have more favorable bonding interactions with CO₂R intermediates. This is consistent with the order of M–CO₂, M–COOH, and M–CO bond lengths that decrease in the following order: Sc–R > Ti–R > V–R for the early 3d MN₄Cs (see the Supporting Information for details). The distinct bonding interactions between the intermediates and different metal centers also results in these three MNCs having different RDSs based on their relative intermediate energies. For instance, at $-0.3 V_{\text{SHE}}$, CO₂ adsorption is rate limiting on ScN₄C, *COOH protonation is rate limiting on TiN₄C, and *CO desorption is rate limiting on VN₄C. On ScN₄C, the intermediate energies are insensitive to applied potential, but the reactants, $\text{CO}_{2(\text{aq})} + 2\text{H}_3\text{O}_{(\text{aq})}^+$, become more unstable relative to *CO₂[−] at more reducing biases. Thus, GC-DFT predicts that more reducing biases lower the free energy of the RDS of CO₂ adsorption on ScN₄C to enable faster kinetics. In

contrast, the RDS on TiN₄C shifts from *COOH protonation to *CO desorption at more reducing biases, whereas on VN₄C, *CO desorption is sluggish and rate limiting at all biases and becomes more prohibitive at more reducing potentials. We discuss the *CO desorption energy trends in more detail for the rest of the MN₄Cs below to highlight its importance in predicting CO₂R performance over all 10 pyridinic 3d-MN₄Cs using GC-DFT.

GC-DFT predicts that *CO₂[−] is not thermodynamically stable on ScN₄C relative to CO_{2(aq)} at biases less reducing than $-1.2 V_{\text{SHE}}$. This suggests that *CO₂[−] is a short-lived species that is stabilized by a proton transfer to form *COOH. Alternatively, *COOH more likely forms through a PCET mechanism with CO_{2(aq)} at the electrified interface that avoids the high energy *CO₂[−] intermediate. As mentioned above, the energy of formation of *COOH from aqueous CO₂ on ScN₄C changes significantly as we sweep the potential from -0.3 to $-1.2 V_{\text{SHE}}$. At $-0.3 V_{\text{SHE}}$, the required thermodynamic energy to form *COOH is +0.90 eV. This difference decreases to +0.17 and -0.76 eV at -0.7 and $-1.2 V_{\text{SHE}}$, respectively, indicating a thermodynamically more facile pathway for *COOH formation at more cathodic potentials. Based on our thermodynamic criteria to assess CO₂R capability, ScN₄C would be able to reduce CO₂ to CO slowly at $-0.7 V_{\text{SHE}}$, but more rapidly at $-1.2 V_{\text{SHE}}$.

CO₂R on Middle 3d Metal MN₄Cs. CO₂R catalysis by Cr-, Mn-, Fe-, and CoN₄C all exhibit modest or favorable thermodynamics for *CO formation, but are inhibited by large energetic penalties for CO desorption. At $-0.3 V_{\text{SHE}}$, the first step of CO₂R over Cr- and MnN₄C proceeds through a PCET to immediately form *COOH as the grand free energy of the second state, *CO₂[−] + 2H₃O⁺, is +0.34 and +0.48 eV higher than that of the first and third states over CrN₄C and +0.36 and +0.41 eV higher over MnN₄C, respectively. Although the third state is nearly thermoneutral with the first one for these species at $-0.3 V_{\text{SHE}}$, its immediate protonation to form *CO is thermodynamically favorable. At more reducing potentials, the formation of *CO₂[−] and *COOH become more thermodynamically accessible, further facilitating the production of *CO; however, CO desorption becomes even more prohibitive at these potentials. This behavior is consistent with experimental observations that MnN_xC produces CO more selectively at lower potentials and very sluggishly at moderate to high cathodic potentials.^{12,19,51,52}

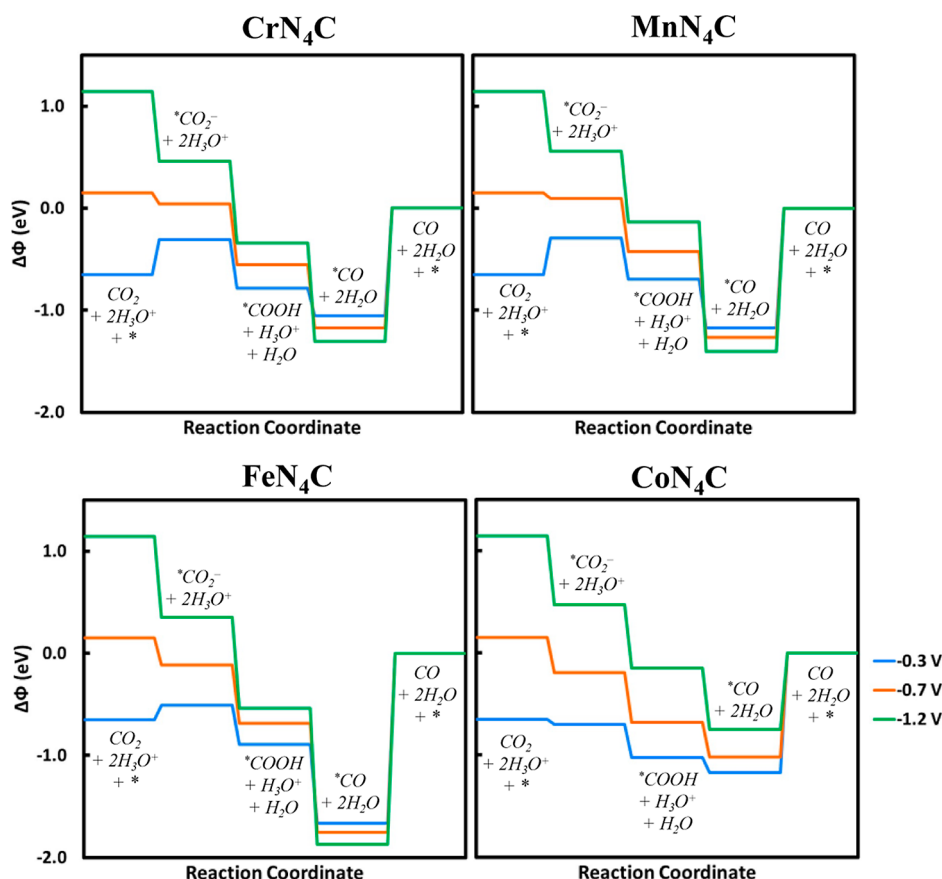


Figure 4. CO₂R reaction coordinate diagrams for Cr-, Mn-, Fe-, and CoN₄C at $\varphi = -0.3$ (blue), -0.7 (orange), and -1.2 V_{SHE} (green). FeN₄C has the potential to produce beyond CO products due to its extremely stable *CO intermediate.

CO₂R catalyzed by Fe- and CoN₄C at -0.3 V_{SHE} could proceed through either a PCET or DPET to form *COOH as the first and second states are nearly thermoneutral. Our thermodynamic analysis based on GC-DFT-computed grand free energies indicates that CO₂R on Fe- and CoN₄C via both DPET and PCET mechanisms is independent of pH, as the *CO₂[−] adsorption and the CO desorption steps do not require the transfer of a proton. On the other hand, a noticeable shift occurs in the reaction mechanism for all four middle MN₄Cs at more reducing potentials where *CO₂[−] becomes substantially more stable than CO_{2(aq)} and the mechanism becomes exergonic until *CO desorption. For biases where $\varphi \geq -0.7$ V_{SHE}, CO₂ is considerably more likely to adsorb on the middle 3d MN₄Cs than to undergo hydrogenation by PCET to form *COOH, which is consistent with commonly observed behavior over homogeneous catalysts.^{46,53,54} Thus, GC-DFT predicts that CO₂R activation by these catalysts is not pH-dependent under moderately cathodic conditions.

As the metal atomic number increases from Cr → Mn → Fe, *CO desorption becomes less favorable. At -1.2 V_{SHE}, the *CO desorption energy, $E_{\text{des}}(\text{CO})$, is 0.567 eV less favorable on FeN₄C than on CrN₄C. It is worth noting that the energetic differences in *CO stability calculated by GC-DFT are not equivalent to the changes in applied bias, as assumed in CHE calculations. This is because GC-DFT directly describes changes in the number of electrons of the quantum mechanically modeled portion of the electrified interface to maintain equilibrium with the Fermi level of the electron reservoir.²⁸

In contrast to the aforementioned MN₄Cs, *CO becomes significantly less stable and thus desorbs more favorably on CoN₄C as more reducing potentials are applied. However, our results predict that highly endergonic *CO desorption slows or poisons CO₂R over CrN₄C, MnN₄C, and FeN₄C at all considered potentials. The large endergonicity of *CO desorption likely causes it to be the RDS of CO₂R to CO for pyridinic CrN₄C, MnN₄C, and FeN₄C, which supports experimental observations of pH-independent CO production over FeN_xC and molecular MNC analogues.^{13,55} While CrN₄C is likely to exhibit less poisoning than Mn- and FeN₄C, experimental studies have found HER to outcompete CO₂R over CrN₄C, which can be partially attributed to the large barrier for CO desorption.⁵¹ However, our results predict that highly endergonic *CO desorption poisons or slows CO₂R over Cr-, Mn-, Fe-, and CoN₄C at all considered potentials. The large endergonicity of *CO desorption likely causes it to be the RDS of CO₂R to CO for pyridinic CrN₄C, MnN₄C, and FeN₄C, which supports experimental observations of pH-independent CO production over FeN_xC and molecular MNC analogues.^{13,55}

The exceptionally large energy required to desorb *CO from FeN₄C at all potentials considered in this study (>1.67 eV) indicates that it would remain on the surface sufficiently long to possibly undergo further conversion to hydrocarbon products. CH₄ production over FeN_xC has been observed experimentally, one of the few known catalysts outside of copper to do so, which has contributed to the intense interest in MNCs for CO₂R.^{9,56} However, the size of the barrier required to desorb *CO from FeN₄C leads us to conclude that

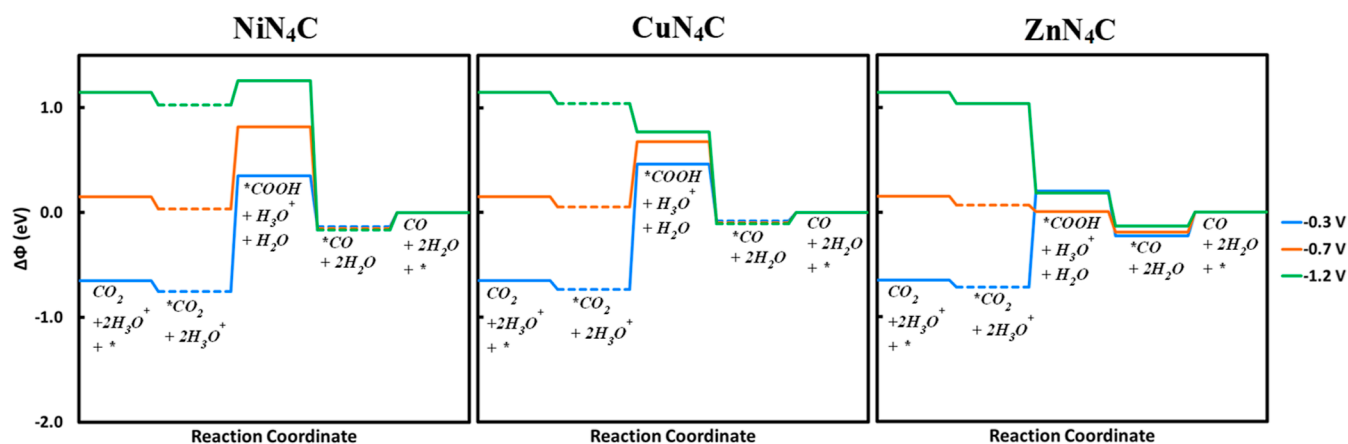


Figure 5. CO₂R reaction coordinate diagrams for Ni-, Cu-, and ZnN₄C at $\varphi = -0.3$ (blue), -0.7 (orange), and -1.2 V_{SHE} (green). Physisorbed molecules are indicated by a dashed line. Note that Ni- and CuN₄C do not chemisorb CO₂ and CO at any considered bias.

pyridinic FeN₄C is likely not the most active FeN_xC motif. A recent study on FeN_xC suggested that the axial coordination of a pyrrolic N ligand to the FeN₄C motif reduces the electron density over the metal center, which decreases the π -backdonation to CO₂ and Fe–C bond strength.³⁰ Similar suggestions have been offered about MnN_xC catalysts,^{19,51} and GC-DFT modeling of CO₂R over MnN_xC and FeN_xC sites with different nitrogen coordinations and functionalities can help determine the nature of the MNC active site responsible for CO₂R on Mn- and Fe-based MNCs.⁵²

For the case of CoN₄C, GC-DFT predicts that the formation of all intermediates at all biases is exergonic relative to the reactants such that CO₂R is thermodynamically accessible at all applied potentials considered, although it is most favored at -1.2 V_{SHE} due to the lowered $E_{\text{des}}(\text{CO})$. This agrees with experimental studies;^{51,55} however, both studies reported intense HER competition with CO₂R over synthesized CoN_xC that was not observed for cobalt-based molecular electrocatalysts.^{57–60} Discrete CoN_xC catalysts have a different coordination sphere that involves pyrrolic nitrogen atoms that affect the electronic density on the Co center, possibly suppressing HER. Therefore, future computational studies that examine the effect of nitrogen coordination may elucidate the mechanism of HER suppression over CoN_xC.

CO₂R on Late 3d Metal MN₄Cs. The most notable difference between the late and middle MN₄Cs is the significantly decreased barrier for *CO desorption. Experimental studies have reported that both Ni- and ZnN₄C are highly selective for CO, which is supported by $E_{\text{des}}(\text{CO})$ being no larger than 0.23 eV. This diminished barrier implies that *CO and CO_(aq) coexist in equilibrium, as shown previously.⁶¹

Similar to FeN_xC, NiN_xC has also generated considerable interest among researchers studying MNC materials. Due to the fact that GC-DFT predicts that CO_{2(aq)} only physisorbs at the NiN₄C surface, it is expected that *COOH must be formed through a PCET over NiN₄C at all potentials. However, as shown in Figure 5, the high energies of the *COOH intermediate relative to CO_{2(aq)} on NiN_xC predict that CO₂R over NiN₄C is not thermodynamically favored at potentials less reducing than -0.7 V_{SHE}. At -0.7 V_{SHE}, the formation of *COOH is still slightly higher in energy than CO_{2(aq)}, so the reaction is expected to be kinetically slow and dependent upon available proton donors. Furthermore, the presence of hydrogen bonding has been shown to stabilize

CO₂R intermediates on NiN_xC.⁶⁵ We found that including one explicit H₂O molecule oriented toward the hydrogen atom of *COOH moderately stabilizes *COOH, but that this effect diminishes at more reducing potentials (see Supporting Information, Figure S10). As shown in Supporting Information, Figure S10, at -0.7 V_{SHE}, the energy of *COOH on NiN₄C is 0.31 eV lower when an explicit H₂O H-bonded to *COOH is added to the model. This makes the overall pathway appear to lie within the limits of our thermodynamic criterion of 0.7 eV. However, in the same paper,⁶⁵ the authors found that the kinetic barrier for the formation of *COOH (including H-bonding) is >1 eV and as such, the NiN₄C site is unlikely to be sufficiently active at 298 K for CO₂R to proceed at an appreciable rate at the considered potentials.

These results contradict previous DFT studies that argued that NiN₄C is the most active and selective site for CO₂R to CO of a synthesized NiN_xC catalyst at moderate potentials.^{15,31,62} However, our results join a growing number of studies that question whether pyridinic NiN₄C is the most active site for CO₂R at moderate potentials as recent studies have reported increased stabilization of the *COOH intermediate as the number of coordinating nitrogen atoms is reduced.^{18,48,63–65} Hossain et al.⁶³ applied a grand canonical approach to study the kinetics of CO₂R on pyridinic NiN₂C, NiN₃C, and NiN₄C, in which they conclude that the NiN₄C site is highly selective toward CO only at potentials more negative than -1.4 V_{SHE}, which is consistent with the results reported herein. In contrast, they predict that NiN₂C is highly selective at potentials between -1.1 and -1.3 V_{SHE}. In an ab initio molecular dynamics simulation that included hydrogen bonding and explicit surface charge, NiNC was found to be the most active and selective Ni-based MNC for CO₂R at -0.65 V_{RHE}.⁶⁶ The rationale given for NiNC's superior performance was that its optimal charge capacity of an excess of two electrons at the studied potential facilitated electrochemical steps without increasing the bond strength of *CO so much as to prevent its desorption. Experimental electrochemical studies of CO₂R on NiN_xC show that CO is produced at potentials ranging from -0.6 to -1.6 V_{SHE}, with the maximum FE of CO lying between -0.9 and -1.3 V_{SHE}.^{14,15,62} These observations and our calculations, which show that *COOH lies prohibitively high in energy, led us to conclude that NiN₄C is not the active site responsible for CO production at less reducing potentials and that NiN_xC sites with $x \leq 3$ are likely

responsible for NiN_xC 's activation of CO_2R to CO with high selectivities, especially at practical potentials that are most relevant to industrial applications (i.e., operating potentials less negative than $-1.0 \text{ V}_{\text{SHE}}$).

While bulk metallic copper is arguably the most studied catalyst for CO_2R , CuN_4C has been studied comparatively less. Our results, shown in Figure 5, predict that CO_2R over CuN_4C is thermodynamically accessible except at $-0.3 \text{ V}_{\text{SHE}}$, where the thermodynamic barrier for $^*\text{CO}_2^- \rightarrow ^*\text{COOH}$ of 1.2 eV is too large to meet our criterion to be consistent with a reasonable reaction rate. Prior experimental work has shown that CuN_xC reduces CO_2 to CO ,¹² albeit with low selectivities. However, the authors found evidence that CuN_xC is not stable and may form Cu nanoparticles that could activate the observed conversion of CO_2 into CO and other hydrocarbons. If clean CuN_4C is successfully synthesized, our GC-DFT results predict that it involves minimal thermodynamic barriers to catalyze CO_2R at ambient temperatures.

Although the stability of the ZnN_4C complex has been questioned,² recent experiments demonstrated that it is a highly active, selective, and robust catalyst for CO_2R to CO . Chen et al.¹¹ and Yang et al.¹⁰ reported 91 and 95% FE of CO on ZnN_xC with maximum FE and CO partial current densities attained at potentials of approximately -0.9 and $-1.2 \text{ V}_{\text{SHE}}$, respectively. Our GC-DFT results predict that for CO_2R on ZnN_4C at -0.3 and $-0.7 \text{ V}_{\text{SHE}}$, $^*\text{CO}_2^-$ is unable to chemisorb and CO_2R instead proceeds via PCET. At $-0.7 \text{ V}_{\text{SHE}}$, the reaction steps are nearly thermoneutral from reactants to products, but become exergonic at the highly reducing potential of $-1.2 \text{ V}_{\text{SHE}}$ with the exception of $^*\text{CO}$ desorption, which is largely invariant to potential and slightly endergonic.

The results discussed above demonstrate the power of GC-DFT for modeling processes at electrified interfaces relative to conventional DFT methods that fix the electron number. For example, GC-DFT clearly shows that a more reducing voltage does not always stabilize adsorbates. In some cases, such as $^*\text{CO}_2^-$ adsorbed to all MN_4Cs , reducing potentials are entirely stabilizing (more favorable adsorption energies), whereas in others, such as $^*\text{CO}$ on FeN_4C , they are destabilizing. In all cases, the effects are unique to both the MNC metal and adsorbate identities. Furthermore, in contrast to the CHE model, directly computing the grand free energies at different biases using GC-DFT shows that the energies generally do not change proportionally to the applied potential. Large differences in explicit-bias effects between some 2D-materials and 3-dimensional metal catalysts were demonstrated by Kim et al.⁶⁶ They found that the chemical reactivity of 2D materials could be considerably more sensitive to applied potential than a Pt bulk metal catalyst, summarily calling into question mechanisms predicted over 2D materials by charge neutral DFT. By comparing the change in the electronic density of states, the authors concluded that the differences arose from the much smaller quantum capacitance of 2D materials, which resulted in comparatively smaller changes in electron number required to occupy or vacate the electronic bands to shift the Fermi level to match the applied potential.

The potential causes non-linear effects not only on grand free energies but also on the projected density of states (PDOS) where the shifts of the energies of the electronic states (electronic orbitals) differ from those predicted by the rigid band model (RBM), such as the CHE approximation. In contrast, a RBM predicts linear stabilizing shifts based on the number of transferred electrons for cathodic reactions. Figure 6

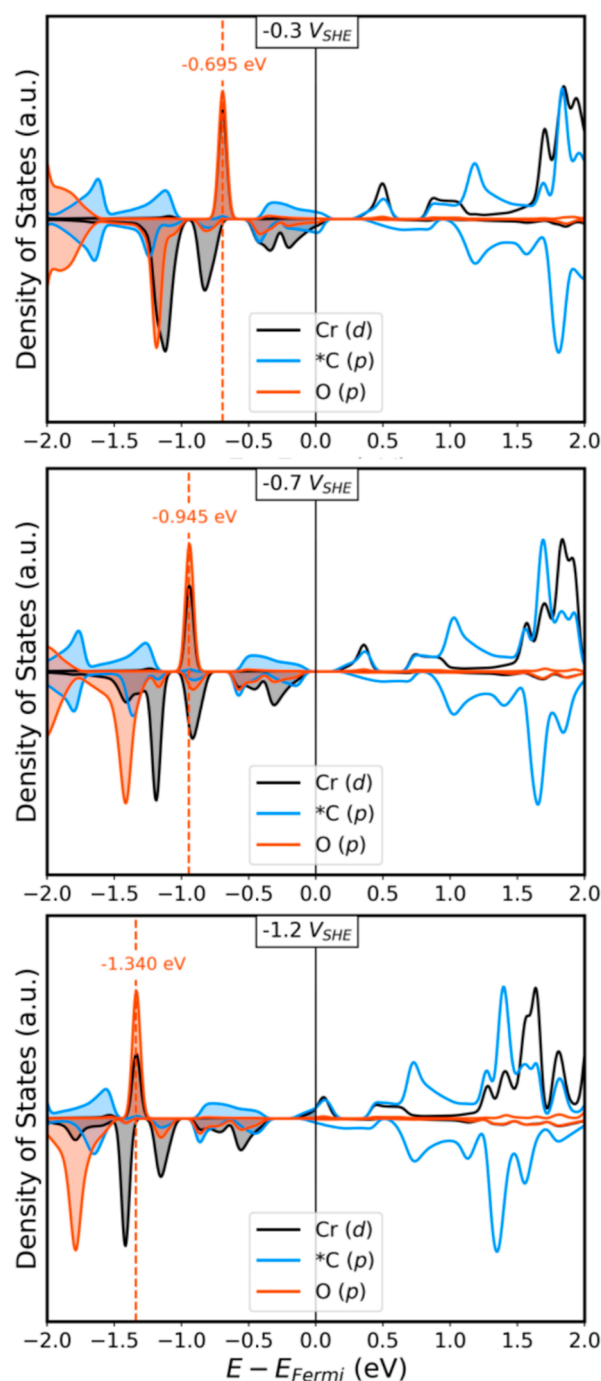


Figure 6. PDOS plots of $^*\text{CO}_2^-$ over CrN_4C at $\varphi = -0.3, -0.7$, and $-1.2 \text{ V}_{\text{SHE}}$. Only the chromium d -states (black), carbon p -states (blue), and oxygen p -states (orange) are included here. The dashed orange line tracks the $^*\text{CO}_2^- \pi^*$ states. All states have been scaled to their highest respective peak values. Note the non-linear shifts and the changing shapes of the states as the applied potential shifts to more cathodic values.

shows the non-linear change in the PDOS computed using GC-DFT for $^*\text{CO}_2^-$ on CrN_4C as a function of potential. For instance, the overlapping O 2p and Cr 3d peaks, initially at -0.695 eV for the $-0.3 \text{ V}_{\text{SHE}}$ case, shift together by 0.25 eV to lower energies relative to the Fermi level, whereas the spin-down Cr 3d peak at -0.820 eV shift only by 0.095 eV when sweeping the applied potential from -0.3 to $-0.7 \text{ V}_{\text{SHE}}$. Furthermore, the initially broad Cr 3d peak at -1.140 eV is

deconvolved by the bias into 2 peaks, including a sharp Cr 3d peak, which shifts together with the spin-down Cr 3d peak that was initially at -0.820 eV, and a Cr 3d peak that shifts together with an O 2p peak, originally overlapped with the shoulder of the broad Cr 3d peak. These effects of the applied bias on both the shapes and peak positions of the PDOS predicted by GC-DFT influence the occupation of states and deviates from those predicted by RBMs, which neglect electronic interactions and assume that the Fermi level sweeps up or down through the DOS by an amount equal to that of the applied potential. Consequently, RBMs do not correctly describe the stabilization/destabilization and change in occupation of states caused by the non-linear potential effect on the PDOS nor its effect on adsorbate geometry, which also influences the degree of orbital mixing.

Table 1 details how the GC-DFT-computed $^*\text{CO}_2^- \pi^*$ peak energies shift within the studied potential range and how they

Table 1. Grand Free Energies of the $^*\text{CO}_2 \pi^*$ State as Computed Using GC-DFT and as Calculated Using the Rigid-Band Model^a

applied potential (V_{SHE})	$^*\text{CO}_2^- \pi^*$ energy (GC-DFT, eV)	$^*\text{CO}_2^- \pi^*$ energy (rigid shift, eV)
-0.3	-0.695	-0.695^b
-0.7	-0.945	-1.095^c
-1.2	-1.340	-1.595^c

^aEnergies are referenced to their respective Fermi energies at each applied potential. The CHE approximation is applied from the reference energy of the $^*\text{CO}_2 \pi^*$ state at $\varphi = -0.3 V_{\text{SHE}}$, which is -0.63 eV. ^bThis value is computed using GC-DFT at $-0.3 V_{\text{SHE}}$ and is expected to be different when computed using conventional DFT methods. ^cShifted values from the $^*\text{CO}_2^- \pi^*$ energy at $\varphi = -0.3 V_{\text{SHE}}$ depending on the number of transferred electrons.

differ from the shifts predicted by the RBM. The details of energy calculations are described in the Supporting Information. Additionally, a comparison of the projected DOS of $^*\text{CO}$ on ScN_4C and VN_4C is presented in Figure S9 of the Supporting Information to address differences of how adsorption energies are affected by applied external bias. These results suggest that future work should explore the nature of the effect of potential on the electronic structure of electrocatalysts using methods such as GC-DFT that fundamentally describe electrified interfaces.

CONCLUSIONS

We applied GC-DFT to investigate the effects of applied potential on the CO_2R energetics and electronic structures of the 10 3d-block transition metal MN_4Cs at applied potentials of -0.3 , -0.7 , and $-1.2 V_{\text{SHE}}$. GC-DFT computed reaction coordinate diagrams exhibit a high sensitivity of the $^*\text{CO}_2^-$ adsorption grand free energy to potential for all MN_4Cs , indicating that the stability of the $^*\text{CO}_2^-$ intermediate is crucial for activating the CO_2R pathway. PDOS plots suggest that metal d-orbital and CO_2 electronic states shift nonlinearly with increasingly reducing applied potentials. This observation provides new fundamental insights into the effects of applied potential on the electronic structure of electrocatalysts. Of the pyridinic MN_4Cs examined, GC-DFT predicts that Sc-, Ti-, Co-, Cu-, and ZnN_4C are active for electrocatalytic CO_2R to CO at moderate to highly reducing potentials (i.e., $\varphi = -0.7$ to $-1.2 V_{\text{SHE}}$). Pyridinic ZnN_4C , specifically, is predicted to possess highly favorable CO_2R thermodynamics at both low

and moderate reducing biases, suggesting that it is a particularly promising candidate for CO_2R to CO. GC-DFT also predicts that CO_2R over CoN_4C is pH-independent at all potentials, which is analogous to the behavior of the molecular cobalt protoporphyrin catalyst. For the cases of pyridinic Cr-, Mn-, and FeN_4C , the rate of CO_2R is predicted to be limited by the prohibitively large grand free energy to desorb CO. This prediction is consistent with prior experimental observations for CO_2R on FeN_xC where the CO formation rate was observed to be pH independent at more negative biases, as the rate limiting CO desorption step does not require the presence of protons.

We suggest that future computational studies investigating CO_2R catalyzed electrochemically by MN_xCs focus on examining the RDS, the dependence of transition state energies to the applied potential, and the effects of N functionality and coordination on the reactivity. The utility of GC-DFT as a computational tool is exemplified by its exceptional ability to predict potential dependent properties and, in contrast to the CHE approach, to distinguish between steps that do not involve proton transfers at varying biases. GC-DFT's broad applicability, more sound fundamental basis, and few limiting assumptions suggest that it is an appropriate method for describing processes at electrified interfaces and thus opens a new horizon for computational electrochemical modeling.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.2c01832>.

Rigid shift (CHE) calculation details, additional MN_xC configurations, solvation details, GC-DFT details, energy calculations, CO_2 adsorption geometries on MnN_4C , bond lengths and angles, comparison of CHE to GC-DFT, PDOS plots of $^*\text{CO}$ on ScN_4C and VN_4C , and effect of hydrogen bonding on $^*\text{COOH-NiN}_4\text{C}$ (PDF)

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Notes

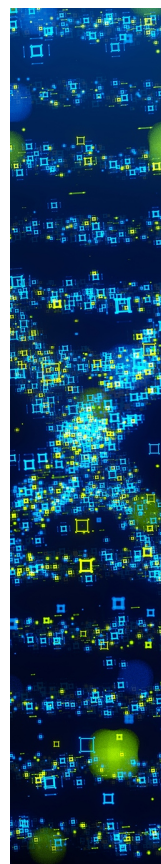
The authors declare no competing financial interest.

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